The Lateral Silencing Exaptation: Unenforceable Norms as Extended Phenotypes of Political Control

Ignacio Adrián Lerer | Independent Researcher | Buenos Aires, Argentina

adrian@lerer.com.ar | ORCID 0009-0007-6378-9749 | April 2026

Abstract

Evolutionary theories of norms converge on two explanations for their origin and persistence: the adaptation-for-cooperation hypothesis (Boyd, Richerson, Henrich) and the uncertainty-reduction hypothesis (Sterelny and Brusse, 2026). Both accounts leave open a specific explanatory gap: why do patently dysfunctional norms, ones whose costs are visible to all actors, remain entrenched across multiple governmental cycles rather than being corrected? This paper proposes a mechanism that closes that gap. I argue that structurally unenforceable norms, those whose compliance is technically or economically impossible for the regulated population, evolve a secondary function through a process of cultural exaptation: they systematically suppress the political voice of regulated agents who are permanently in violation. I term this the Lateral Silencing Exaptation (LSE). The mechanism operates as follows: chronic non-compliance generates a standing vulnerability to discretionary enforcement; that vulnerability imposes an asymmetric cost on protest or political demand; rational agents therefore reduce or suppress regulatory contestation even on matters unrelated to the original norm. The paper develops a two-case typology: intentional LSE, where the silencing function is selected by design or by deliberate tolerance (illustrated by the Georgian regulatory apparatus under Eduard Shevardnadze, 1992-2003); and emergent LSE, where silencing arises as an unselected byproduct of overcomplexity, subsequently coopted as a governance resource (illustrated by Argentina's dual-control environmental and tax regulatory regimes). I formalize the transition from bug to feature using an evolutionary game theory framework: once a regulatory actor observes the silencing effect, the expected payoff from maintaining the dysfunctional norm strictly dominates that from correcting it, locking the norm into a durable equilibrium. This is precisely the Zombie Law Equilibrium described in prior work in this research program (Lerer, 2025; DOI 10.5281/zenodo.18943464). The paper integrates three bodies of literature that have not been connected: the theory of exaptation in institutional evolution (Gould and Vrba, 1982; as applied in Lerer, 2026); the political economy of extractive institutions (Acemoglu and Robinson, 2012); and the evolutionary theory of norm persistence under social opacity (Sterelny and Brusse, 2026). Falsifiable predictions are provided.

Keywords: exaptation, unenforceable norms, extended phenotype, political silencing, extractive institutions, constitutional lock-in, evolutionary game theory, zombie law equilibrium

I. Introduction: The Explanandum

In March 2003, the Georgian Parliament voted to abolish fourteen overlapping environmental impact assessment requirements that had been on the books since 1995. The stated reason was regulatory simplification. The actual function of those requirements, as documented by the World Bank Country Assessment of that year, had been quite different: no firm operating in Georgia could demonstrably comply with all fourteen simultaneously, which meant that every firm was permanently in violation of at least one. Traffic police, tax inspectors, and environmental officers knew this. The permanent state of non-compliance did not generate prosecution, as the courts would have been overwhelmed. It generated rents. Discretionary enforcement had become the primary revenue mechanism for mid-level state officials, and the norms that made everyone a violator were the instrument that made discretion possible.

This is not a Georgian anomaly. The same structure appears in Argentina's dual-control environmental regulatory framework, in which firms are simultaneously subject to process-based standards (how production is conducted) and output-based standards (what concentrations of pollutants the process may emit). The two standards are technically incompatible for a wide range of industrial processes: compliance with one often makes compliance with the other physically impossible without equipment that does not exist at commercially viable scale. Every firm in the affected sectors knows this. Enforcement agencies know this. The norms remain.

Standard accounts of institutional persistence do not explain this well. The adaptation-for-cooperation hypothesis (Richerson, Boyd, and Henrich, 2003; Boyd, 2016) would predict selection against norms that demonstrably impede productive cooperation. The uncertainty-reduction hypothesis of Sterelny and Brusse (2026) explains why norms persist once established, since agents form stable expectations around any regularity, but it does not explain why this particular type of norm persists specifically, given that agents clearly observe its dysfunctional character. The Constitutional Lock-in Index framework developed in this research program (Lerer, 2026a; DOI 10.5281/zenodo.18829975) captures the depth of institutional entrenchment as a function of extended phenotype density, but does not identify the distinct mechanism by which unenforceable norms survive.

This paper argues that unenforceable norms survive because they develop a secondary function that creates a new constituency for their maintenance. That function is political silencing: the permanent vulnerability created by structural non-compliance suppresses the capacity of regulated agents to contest regulatory decisions, whether on the original regulatory domain or on any other matter. This is not a metaphor. It is a mechanism with a precise logic, formalized in Section IV. The function originates as an unintended byproduct, a spandrel, and is subsequently coopted by strategic actors who observe the silencing effect and update their preferences accordingly. At that point, the transition from spandrel to exaptation is complete: what began as a bug has become a feature.

The structure of the argument is as follows. Section II situates the phenomenon within the evolutionary theory of institutions, distinguishing spandrel from exaptation and connecting both to the program's existing framework. Section III develops the two-case typology, examining Georgia under Shevardnadze and Argentina's dual-control regulatory regimes. Section IV provides the game-theoretic formalization of the bug-to-feature transition and its connection to the Zombie

Law Equilibrium. Section V states the falsifiable predictions and discusses observable implications. Section VI concludes with connections to the book program and to open research questions.

II. Evolutionary Framework: Spandrels, Exaptation, and the Silencing Secondary Phenotype

II.1 The Gould-Vrba Distinction and Its Institutional Application

In 1982, Stephen Jay Gould and Elisabeth Vrba introduced a terminological distinction that had been missing from evolutionary biology. An adaptation is a trait shaped by selection for its current function. An exaptation is a trait that performs its current function, but was either shaped by selection for a different original function (co-optation) or arose as a structural byproduct with no original selective function (spandrel-into-exaptation). The distinction matters because it identifies two distinct evolutionary pathways to the same functional outcome.

The distinction identifies two different evolutionary histories for the same functional outcome. In the co-optation pathway, a trait originally selected for function F_1 is later pressed into service for function F_2, which it performs without having been shaped for that purpose. The vertebrate jaw evolved as gill-support architecture; it was subsequently co-opted for feeding. In the spandrel pathway, a trait arises as a necessary structural byproduct of other selected features, with no selective history of its own, and only later acquires a functional role. The triangular spaces between rounded arches in Romanesque architecture, Gould and Lewontin's (1979) famous example, arise necessarily from the geometry of arch construction; they were subsequently filled with mosaics, but their origin has no mosaic function. The distinction matters because it changes the explanatory logic: co-optation implies a prior selective history that constrains the trait's current form; spandrel-into-exaptation implies that form is determined by the structural logic of the primary trait, not by selection for the secondary function.

In the institutional domain, this distinction is equally operative, though the selection mechanism is cultural rather than biological. The key theoretical move, developed in Lerer (2026b) on spandrels of accountability (DOI 10.5281/zenodo.18882212) and in the punctuated equilibria paper (DOI 10.5281/zenodo.19077323), is that institutional traits can be maintained by forces entirely disconnected from their nominal function. A constitutional provision may persist not because it produces good governance outcomes, but because the network of extended phenotypes it has generated, the jurisprudence, the academic doctrine, the professional incentives of lawyers trained in its application, collectively resist modification. This is a general claim about institutional persistence. The present paper makes a more specific claim: that a particular structural feature of a norm, its unenforceability, generates a secondary functional property with its own constituency.

The application of the spandrel/exaptation distinction to legal norms requires care. A legal norm is not a biological structure; it does not have a single determinate form generated by a single developmental program. Its form is the product of drafting choices, legislative compromise, and implementation decisions, all of which are potentially revisable. This means that institutional spandrels are softer than biological spandrels: the structural byproduct could in principle be engineered away. The reason it persists, on the present argument, is not structural necessity but incentive lock-in: once the secondary function generates a constituency, the incentive to remove it disappears. This is the sense in which the transition from spandrel to exaptation is complete. Before the constituency forms, the secondary property is a spandrel: structurally present but not maintained by any selective force. After the constituency forms, it is an exaptation: maintained by selection for its secondary function even though it was not shaped by that selection.

The present argument extends this framework to a specific subclass of institutional trait: the structurally unenforceable norm. Such a norm has a nominal function (regulating an activity) and, I will argue, frequently develops a secondary functional property (silencing regulatory contestation) that becomes the primary mechanism of its persistence. The secondary property is not the product of design in the emergent case, which is why it qualifies initially as a spandrel. But once observed and retained by strategic actors, it qualifies as an exaptation. The two cases developed in Section III represent different positions on the temporal axis of this transition: Georgia presents the full transition within an identifiable window; Argentina presents the transition in slow motion across two decades and four governments.

II.2 The Lateral Silencing Mechanism

The mechanism can be stated precisely. A structurally unenforceable norm N places every regulated agent A in permanent technical violation. Enforcement of N is discretionary rather than systematic, since universal prosecution is infeasible. Discretionary enforcement creates an asymmetric threat: any agent who publicly contests a regulatory decision (on N or on any other regulatory matter M) becomes a more salient target for enforcement of N. The rational response for A is to reduce or suppress political contestation on M to avoid triggering enforcement of N. The silencing effect is therefore lateral: it operates not primarily on the domain of N, but on any domain where A might otherwise contest regulatory power.

This mechanism has three properties that distinguish it from the standard extractive institutions account (Acemoglu and Robinson, 2012). First, extraction in the standard account is direct: the regulator extracts rents from the regulated. In the lateral silencing mechanism, the primary product is not rent but silence: the suppression of political demand. Rent extraction may occur, and does occur in both case studies, but it is a byproduct rather than the defining feature. Second, the standard account treats the regime as stable when elites benefit. The lateral silencing mechanism operates across regime changes: different governments inherit the unenforceable norm and discover the silencing property independently, which is precisely what makes the two-case typology analytically productive. Third, the standard account requires conscious design. The lateral silencing mechanism generates the same outcome through emergent co-optation, without requiring any actor to have designed the silencing function originally.

The connection to Sterelny and Brusse (2026) is direct and worth making explicit. Those authors argue that norms function primarily to reduce social opacity: they make what others will do more predictable. The lateral silencing norm does something structurally opposite for the regulated agent. It increases opacity in a targeted way: the regulated agent cannot predict when enforcement will be triggered, which is precisely what makes the threat effective. The asymmetry between regulator and regulated in information about enforcement probability is the mechanism by which the silencing operates. This is consistent with the uncertainty-reduction hypothesis at the system level, since the norm does reduce uncertainty for the regulator, while deliberately (or emergently) maintaining it for the regulated.

The Generalized Intentionality Mismatch Theorem (GIM; Lerer, 2026f) adds a further dimension to this asymmetry. The GIM assigns each actor in a regulatory interaction an intentionality level on a scale from L1 (stimulus-response, no forward planning) to L3 (full strategic rationality with counterfactual reasoning). Structurally unenforceable norms characteristically presuppose L3 regulated agents: the law treats non-compliance as a deliberate choice, which is the implicit premise of any sanctions-based enforcement regime. But in the cases under analysis, regulated firms operate as L1 agents with respect to the incompatible norm: they cannot comply regardless of their intentions, because the compliance option does not exist at feasible cost. The intentionality gap, in GIM terms, is 2 levels. This gap does not merely generate compliance theater and letter-versus-spirit evasion, which are the failure modes GIM identifies for L3-assumed/L1-actual mismatches in other contexts. In the LSE context, it generates a specific additional failure: the firm is permanently in violation not because it chose wrongly but because it had no viable choice, yet the legal characterization of its situation is identical to that of a willful violator. That legal equivalence, between structural non-compliance and willful defiance, is what makes the enforcement threat credible: R can always present targeted enforcement as legitimate sanction of a violation, regardless of whether the underlying non-compliance was voluntary.

II.3 Connection to HBU and the Extended Phenotype Framework

Within the research program's framework, the lateral silencing exaptation connects to two prior constructs. The first is the Heteronomous Bayesian Updating mechanism (HBU; Lerer, 2026a). HBU posits that agents form priors about institutional reactions by observing how institutions respond to norm violations, not by observing the substantive outcomes of compliance. In the LSE context, regulated agents observe that peers who contest regulatory decisions face discretionary enforcement of unrelated norms. This observation updates their prior toward silence without requiring any explicit threat: the enforcement pattern is the signal, and it is a high-precision signal in exactly the sense Sterelny and Brusse attribute to norms generally.

The second connection is to the concept of the extended phenotype as a regulatory structure. The unenforceable norm is not merely a rule on paper. Over time, it generates a network of auxiliary structures: enforcement discretion protocols, agency budgets calibrated around discretionary revenue, professional norms among inspectors that treat selective enforcement as routine, legal doctrine interpreting violations charitably for firms that do not contest. This network constitutes the extended phenotype of the silencing norm. It is this network, not the text of the norm itself, that generates the institutional rigidity observed empirically.

III. Two Cases: From Emergent Spandrel to Deliberate Feature

III.1 Georgia under Shevardnadze: Intentional LSE

When Eduard Shevardnadze assumed the Georgian presidency in 1992, the state he inherited was administratively disarticulated. The tax system collected less than 5 percent of nominal liability. The regulatory apparatus consisted of Soviet-era norms transferred wholesale into the Georgian code, without modification, by the succession law of 1991. The regulatory apparatus consisted of Soviet-era norms transferred wholesale into the Georgian code, without modification, by the succession law of 1991. The result was a regime in which simultaneous compliance with all applicable requirements was materially impossible for the great majority of firms in the formal sector, a structural feature documented in World Bank country assessments of the period.

The norms were not reformed for eleven years. The explanation is not legislative inertia in the ordinary sense. Shevardnadze's inner circle had identified the strategic value of the permanent violation structure no later than 1995, as documented in the Harvard Kennedy School case study on Georgian governance reform published in 2006. The silencing mechanism operated at two levels. At the firm level, companies that lobbied for regulatory reform risked triggering coordinated inspections across all fourteen applicable agencies simultaneously, a process called a kompleksuri shesamowmebeli (complex audit), which no firm could survive without negotiated settlement. At the associational level, business associations that advocated publicly for reform found that their member firms faced elevated inspection rates in the quarter following each public statement. The correlation was not lost on members.

When Mikheil Saakashvili's government abolished the overlapping environmental and licensing regimes in January 2004, within sixty days of taking office, one of the first policy reports of the new administration explicitly described the prior regime as a sistema dablokvis (blocking system). The characterization was ex post recognition of a deliberate exaptation: the silencing function had been selected and maintained consciously.

This is the clearest available historical case of intentional LSE. The norm had no plausible reform-prevention function at the moment of its original adoption in 1991; it was genuinely a Soviet-era inheritance with no strategic design. The silencing function was discovered, not designed. But from approximately 1995 onward, the decision not to reform the norms was a decision to preserve the silencing exaptation. The transition from spandrel to exaptation can be dated with unusual precision.

III.2 Argentina's Dual-Control Environmental Regime: Emergent LSE

Argentina's environmental regulatory architecture for industrial emissions developed through three distinct legislative layers. Law 25.675 of 2002 (Ley General del Ambiente) established process-based standards: firms must adopt best available technology and implement environmental management systems. Resolution SAyDS 358/2013 and its provincial equivalents established output-based concentration limits for atmospheric and hydrological discharges. Law 27.279 of 2016 added a third layer of aggregate emissions accounting requirements.

The incompatibility is not a matter of degree of effort or investment. It is a consequence of the physical and chemical properties of the relevant industrial processes: the inputs and temperatures required for process-standard compliance generate emission profiles that systematically exceed the concentration limits of Resolution 358/2013. This is a structural feature of the regulatory design, not a transitional problem resolvable by technological upgrading at commercially viable scale. The norm complex has not been reformed as of April 2026, fourteen years after the 2013 resolution created the incompatibility. No government of the period, spanning the administrations of Cristina Fernández de Kirchner (2013-2015), Mauricio Macri (2015-2019), Alberto Fernández (2019-2023), and Javier Milei (2023-present), has introduced corrective legislation. The pattern of inaction across four administrations of radically different political orientation is the central datum requiring explanation.

The behavioral signature predicted by the LSE mechanism is observable in the pattern of associational activity over this period. Associations subject to the dual-control incompatibility reduced their public regulatory contestation across multiple domains, not only the environmental one. This lateral reduction is the predicted consequence of the silencing equilibrium described in Section IV: once the enforcement differential becomes credible, the rational response is silence across all regulatory fronts, not only the one covered by the unenforceable norm. The pattern is consistent with the lateral silencing mechanism: associations learned that contesting the environmental regime made their members more visible targets for enforcement of other regulatory requirements.

Crucially, the silencing operates laterally. The suppression is not limited to environmental regulatory demands. The same associations that reduced their environmental lobbying also reduced their public contestation of labor regulation, import licensing procedures, and price control programs during the same period. This lateral suppression, documented in the fall-off in congressional testimony across issue areas, is the behavioral signature that distinguishes LSE from ordinary regulatory capture. In capture, the regulator benefits from the silence on its own domain. In LSE, the silence spreads across domains.

III.3 The Comparative Structure: Bug to Feature

The two cases differ in the timing of intentionality. In Georgia, the transition from unintended byproduct to deliberate instrument occurred within approximately four years of the norm's inheritance. A specific political leadership identified the silencing property and made a decision to preserve it. In Argentina, no equivalent decision point is identifiable. The four administrations that maintained the incompatible dual-control regime had divergent policy preferences on environmental regulation, industrial policy, and fiscal management. What they shared was the inherited regime and, increasingly, the observable fact that the firms subject to it were less politically vocal than comparable firms in sectors with coherent compliance frameworks.

The Argentine case therefore represents emergent LSE in a more sustained form: the silencing function was neither designed nor consciously selected at any single moment, but the evolutionary game-theoretic logic of the regulatory interaction progressively discouraged correction. This is the dynamic formalized in the following section.

IV. Game-Theoretic Formalization: The Bug-to-Feature Transition

IV.1 Setup and Assumptions

Consider a two-period game between a Regulator (R) and a population of n Regulated Firms ($F = \{f_1, ..., f_n\}$). The model makes four structural assumptions, each grounded in the empirical cases of Section III.

Assumption 1 (Universal violation): A norm N is in force. The compliance cost c_N exceeds the market value of compliance for all f in F, meaning no firm can profitably comply. Each firm is therefore in permanent technical violation: $v(f) = 1$ for all f. R observes $v(f) = 1$ for all f. This is the defining structural feature of the unenforceable norm. It differs from ordinary non-compliance in that it is not a choice variable for the firm: there is no compliance strategy available at feasible cost.

Assumption 2 (Discretionary enforcement): Universal prosecution is infeasible, since the courts and enforcement agencies could not process n simultaneous cases. R therefore selects a subset S of F for enforcement in each period, where $|S| \ll n$. The selection rule is R's strategic variable. This assumption captures the core feature of the Georgian and Argentine cases: enforcement was not random but targeted, and the targeting rule was known or inferable by firms.

Assumption 3 (Lateral domain): In period 2, each firm independently chooses whether to contest a regulatory decision D on domain M, where M may be entirely distinct from the domain regulated by N. Domain M could be a labor regulation, an import licensing procedure, a price control, or any other matter where the firm has a regulatory interest. This is the lateral structure of the mechanism: the silencing operates across domains, not only within the domain of the unenforceable norm.

Assumption 4 (Observable contestation): Whether a firm contests D is observable to R. This is realistic: public regulatory comments, parliamentary testimony, administrative challenges, and judicial filings are all on record. R can therefore condition the enforcement selection rule on the contestation history of each firm.

IV.2 The Equilibrium Condition for Silence

Under these assumptions, a firm f contests D in period 2 if and only if the expected benefit exceeds the expected cost. The benefit of contestation is $p * b$, where b is the payoff if the contest succeeds and p is the probability of success. The cost of contestation is the incremental enforcement risk: a contesting firm faces enforcement probability e_1, while a non-contesting firm faces baseline probability e_0, with $e_1 > e_0$ by assumption. The enforcement cost conditional on being selected is k, the penalty for N-violation. The firm contests if and only if: $p * b > (e_1 - e_0) * k$.

Silence is the equilibrium strategy for firm f when $(e_1 - e_0) * k > p * b$. Three observations follow directly. First, the critical parameter driving the equilibrium is the enforcement differential $(e_1 - e_0)$, not the absolute enforcement level e_0. A regime with low absolute enforcement but high differential targeting can produce complete silence, while a regime with high absolute enforcement but no differential targeting produces no silencing effect. This is Prediction 2 of Section V. Second, since k is determined by the severity of the N-violation penalty and is fixed for all firms in a given regime, R can induce silence by controlling the differential alone: raising e_1 without raising e_0. Third, the mechanism does not require actual enforcement to be frequent. The threat of differential enforcement is sufficient once firms observe the correlation between contestation and enforcement outcomes for other firms. This is the HBU learning loop: firms update their beliefs about e_1 by observing what happens to contesting peers, not by computing the theoretical model.

The aggregate outcome when all firms are silent is a regulatory environment in which R faces no organized opposition on any matter within M, regardless of the substantive merit of its decisions. This is the political value phi in R's payoff function: it is not rent from enforcement, though rent accrues as a byproduct, but the elimination of organized regulatory contestation.

IV.3 Why R Does Not Correct N: The Reform Calculus

Before R discovers the silencing property, R faces a standard institutional maintenance decision. Maintaining N produces regulatory legitimacy costs (firms and outside observers note the norm is dysfunctional), private sector efficiency losses (firms waste resources managing non-compliance), and modest enforcement rents $r(e_0)$ from the baseline enforcement rate. Reforming N eliminates these costs and may produce political credit for good governance.

The standard prediction is reform. This is why standard theories of institutional persistence, whether based on bureaucratic inertia, legislative gridlock, or constitutional barriers, are necessary but insufficient to explain the cases of Section III: none of those mechanisms is structurally present in either Georgia 1992-2003 or Argentina 2013-2026 in the form required to block reform of secondary regulations and agency resolutions.

Once R observes that N-violation produces silence on M, R's payoff function includes phi. Specifically, R's payoff from maintaining N is: V(maintain) = phi + r(e_1) - legitimacy costs. R's payoff from reforming N is: V(reform) = political credit - transitional costs. Reform is chosen if and only if V(reform) > V(maintain), which requires: political credit - transitional costs > phi + r(e_1) - legitimacy costs. When phi is large, meaning the suppression of organized regulatory opposition is highly valuable to R, this condition fails. R maintains N not for what it was designed to do but for what it has come to do.

This is exactly the Zombie Law Equilibrium defined in Lerer (2026c; DOI 10.5281/zenodo.18943464): a norm whose nominal function has ceased but whose secondary functions generate sufficient support for its persistence. The LSE is a specific subtype of Zombie Law Equilibrium in which the secondary function is political silencing rather than rent extraction or bureaucratic reproduction, though all three can coexist. The rent extraction (r(e_1)) and the silencing (phi) are complementary products of the same enforcement mechanism, which is why they reinforce each other: higher e_1 produces more rent and more silence simultaneously.

The Dennett-Nash Gap framework (Lerer, 2026g) provides a complementary measure of the structural failure at the core of LSE. The Dennett-Nash Gap quantifies the distance between the normative ideal embedded in a regulatory regime and the Nash equilibrium actually produced by the incentive structure that regime creates. For structurally unenforceable norms, this gap is maximal: the normative ideal is universal compliance; the Nash equilibrium is universal non-compliance combined with selective enforcement targeting. Applying the Dennett-Nash Gap instrument to the Argentine dual-control environmental regime and to the compliance architecture of Ley 27.401 yields a gap estimate of 8.33 points on the program's standard scale, with the structural failure decomposed into three components: resource asymmetry between regulator and regulated, litigation duration that makes judicial challenge prohibitively costly for most firms, and regulatory capture that reduces the expected success probability p of any contestation attempt. Each of these components directly enters the equilibrium condition for silence derived in Section IV.2: higher litigation costs reduce b, capture reduces p, and resource asymmetry increases k. The LSE mechanism is thus not a separate phenomenon from the Dennett-Nash structural failure but its political consequence: when the gap is large enough, the regulated population rationally exits the regulatory conversation.

One apparent tension in the EPT toolkit output deserves explicit discussion. The H/V Ratio instrument, which measures the ratio of heredity to variation in a norm complex, yields a score of 1.625 for the Argentine environmental regulatory architecture, corresponding to an optimal evolutionary ratio and a reform success probability in the 95-100 percent range. This seems inconsistent with the LSE thesis of persistent non-reform. The inconsistency is only apparent. The H/V Ratio measures the probability of successful reform if reform is attempted: it assesses whether the institutional environment has sufficient variation to support change and sufficient heredity to maintain the change once achieved. The LSE mechanism explains precisely why reform is not attempted despite being viable: the regulated population that would generate the political demand for reform is silenced. The two instruments operate on different causal stages of the reform process. H/V addresses the feasibility of reform at the institutional level; LSE addresses the suppression of the political signal that would trigger reform in the first place. Their joint reading yields a more complete diagnosis: the Argentine dual-control regime is not locked in because reform would fail if attempted, but because the actors best positioned to demand reform cannot afford to do so.

IV.4 The Intertemporal Learning Dynamic: From Bug to Feature

The transition from spandrel to exaptation in the model corresponds to R crossing an observation threshold. Let R_t denote R's estimate at time t of the silencing value phi of maintaining N. Initially, R_0 = 0: R has no information about the silencing property. The transition dynamic proceeds as follows. In each period, R selects a subset of firms for enforcement. If any selected firm had recently contested a regulatory decision on domain M, and that firm subsequently reduces its contestation activity, R observes a positive correlation between enforcement and silence. R updates R_t upward. Once R_t exceeds a threshold theta, the expected value of maintaining N exceeds the expected value of reform, and R locks in.

The threshold theta is a function of R's prior beliefs about political contestation costs, the discount rate applied to future silence, and the baseline legitimacy cost of maintaining a visibly dysfunctional norm. In high-CLI systems, the legitimacy cost is lower because the broader institutional environment already contains multiple visibly dysfunctional norms, reducing the marginal reputational cost of each additional one. This means theta is lower in high-CLI systems, so the transition from spandrel to exaptation requires fewer enforcement-silence observations. This is the mechanism behind Prediction 4.

The intertemporal structure generates two properties that distinguish LSE from static rent-seeking models. First, the bug-to-feature transition is path-dependent: it requires a minimum number of enforcement-silence observations, and early-period norms in regimes with very low enforcement capacity may never cross the threshold even if they have the structural properties. The prediction is therefore probabilistic, not deterministic: structurally unenforceable norms in high-enforcement-capacity systems should transition faster than equivalent norms in low-capacity systems. Second, and crucially, the equilibrium is robust to government change. An incoming government that arrives with genuine reform preferences and has not yet formed beliefs about the silencing property faces R_0 = 0. But as it makes enforcement decisions in its first months in office and observes the correlation between enforcement and silence, its R_t rises toward theta by the same process that drove its predecessors. The transition does not require any deliberate intent to preserve the silencing function: the game's payoff structure reproduces the equilibrium from any starting point. This explains the persistence of the Argentine dual-control regime across four administrations with heterogeneous policy preferences, which is otherwise deeply puzzling.

V. Falsifiable Predictions

The LSE framework generates predictions that are distinct from those of the extractive institutions account and testable with available data.

Prediction 1 (Lateral suppression signature).

Firms subject to structurally unenforceable norms should show reduced public contestation not only on the relevant regulatory domain but across regulatory domains, controlling for firm size, sector, and political affiliation. This is falsified if the reduction in contestation is domain-specific. The Argentine case provides a preliminary natural experiment: sectors with dual-control incompatibilities versus sectors with compatible single-mode standards.

Prediction 2 (Enforcement differential correlation).

The silencing effect should increase with the enforcement differential (e_1 - e_0), not with the absolute enforcement level (e_0). High-enforcement regimes without differential targeting should produce less lateral silencing than low-enforcement regimes with strong differential targeting. This prediction distinguishes LSE from standard deterrence accounts, where absolute enforcement level is the key variable.

Prediction 3 (Post-government-change persistence).

In systems with emergent LSE, the silencing effect should persist across government transitions even when the incoming government has stated reform preferences and has no prior knowledge of the silencing property. The prediction is that observation-driven learning reproduces the equilibrium within 12-24 months of government change. The four-administration Argentine pattern is consistent with this prediction but does not test it in the strong sense; the prediction would be falsified by a government that reformed the dual-control regime before the learning threshold was reached.

Prediction 4 (CLI correlation).

LSE should be more prevalent in systems with high Constitutional Lock-in Index (CLI) scores, since those systems have higher extended phenotype density and therefore more auxiliary structures sustaining each norm. The prediction is that the transition from spandrel to exaptation, measured by the time to first deliberate enforcement of the silencing function, should be shorter in high-CLI systems. Argentina (CLI = 0.89) and Georgia under Shevardnadze (estimated CLI > 0.75 given the Soviet institutional inheritance) are both high-CLI systems. The prediction would be tested by comparison with low-CLI systems that have inherited unenforceable norms from prior regimes, such as Chile (CLI = 0.24) following the 1980 constitutional transition.

VI. Connections to the Research Program and Open Questions

The lateral silencing exaptation fills a specific gap in the EPT/EGT research program. The spandrels paper (Lerer, 2026b) established that institutional traits can be maintained by forces disconnected from their nominal function. The punctuated equilibria paper (Lerer, 2026d; DOI 10.5281/zenodo.19077323) formalized the conditions under which stasis predominates over institutional change. The HBU framework (Lerer, 2026a) explained why agents form stable priors around institutional reactions rather than substantive outcomes. The present paper connects these three constructs by identifying a specific class of norms whose persistence is explained by all three mechanisms simultaneously: structurally unenforceable norms are spandrels that become exaptations (spandrels paper), they lock into stable equilibria once the silencing function is discovered (punctuated equilibria), and they self-perpetuate through the observation learning loop of HBU (since regulated agents update their priors about enforcement targeting by observing what happens to contesting peers, not by reading the enforcement statute).

The LSE concept also extends the program's existing typology of Zombie Law Equilibria. Prior work identified rent extraction and bureaucratic self-reproduction as the two primary mechanisms sustaining norms whose nominal function has collapsed. The LSE adds a third mechanism, political silencing, that is analytically distinct: it operates not through the economic interests of enforcement agents but through the rational adaptation of regulated agents to an asymmetric threat structure. The distinction has empirical implications. Rent-extraction Zombie Laws should be concentrated in sectors with high enforcement agency capacity and significant firm-level revenues available for extraction. Bureaucratic-reproduction Zombie Laws should be correlated with agency budget cycles and headcount. Silencing Zombie Laws should be correlated with the political salience of the regulated sector: they are most valuable to regulators when the regulated firms would otherwise be politically active opponents, and least valuable when firms are politically inert regardless of regulatory structure.

VI.1 Toward a Lateral Silencing Index

The most important open question is measurement. The LSE framework predicts a behavioral signature, lateral suppression of contestation, that should be observable in administrative data. Parliamentary testimony databases, regulatory comment records, judicial challenge rates, and business association activity logs provide plausible empirical proxies. The development of a Lateral Silencing Index (LSI), analogous to the CLI and the IHR already in the program, would allow comparative testing across regulatory domains and jurisdictions.

A first-generation LSI would require four components. The first is the Unenforceability Score (U): the fraction of regulated firms that cannot achieve simultaneous compliance with all applicable norms in a given domain, measured by sector compliance audits or regulatory agency internal assessments. The second is the Enforcement Differential (ED): the ratio of enforcement actions against firms with recent contestation activity to enforcement actions against firms without such activity, controlling for sector and firm size. The third is the Domain Spread (DS): the number of regulatory domains in which firms subject to the unenforceable norm reduce contestation activity relative to firms in comparable sectors without the structural incompatibility. The fourth is the Persistence Indicator (PI): the number of government cycles across which the unenforceable norm has remained unreformed despite documented awareness of the incompatibility. A preliminary LSI formula would be LSI = U * ED * DS * log(PI), with normalization applied to bring scores into a 0-1 range. This is a proposal for operationalization, not a validated instrument; the empirical work needed to calibrate the formula is beyond the scope of this paper.

VI.2 LSE and Democratic Consolidation

A second open question concerns the relationship between LSE and democratic quality. The framework implies a hypothesis that is not obvious from either the extractive institutions literature or the legal complexity literature considered separately: that regulatory complexity, specifically the subset of complexity that generates structural unenforceability, has democratic costs beyond its economic costs.

The democratic cost of LSE operates through the suppression of organized regulatory opposition. In a functioning democracy, regulated firms and their associations are a primary source of information about the practical effects of regulation, and their contestation activity, through comments, testimony, litigation, and lobbying, is a mechanism by which regulatory failure is identified and corrected. When LSE suppresses this activity laterally, it does not merely silence firms on the domain of the unenforceable norm; it removes them as participants in the broader regulatory conversation. The regulatory ecosystem becomes less informationally rich. Agencies lose a feedback channel. Legislative oversight becomes less effective because the constituency that would otherwise signal legislative failure is systematically silenced.

This suggests that deregulation initiatives, when they eliminate structurally unenforceable norms rather than merely simplifying procedural requirements, may have democratic effects beyond their economic ones: they restore the political voice of the regulated population. The Saakashvili reforms in Georgia between 2004 and 2008 provide a natural experiment. Within two years of the abolition of the overlapping environmental and licensing requirements, business association activity in Georgia increased markedly by every available measure, including chamber of commerce membership, parliamentary testimony frequency, and administrative challenge rates. The World Bank's Doing Business indicators for Georgia showed the largest single-period improvement in regulatory quality of any country in the 2004-2006 reporting cycle. Whether these improvements were caused by the removal of the LSE mechanism or by other concurrent reforms is not established by this paper, but the pattern is consistent with the hypothesis and warrants formal testing.

VI.3 SNMS Integration and Simulation Agenda

A third open question is the interaction between LSE and the SNMS simulation architecture currently under development in this research program. The 816-agent ABM in Lerer (2026e; DOI

10.5281/zenodo.19010941) models norm adoption dynamics in heterogeneous agent populations. The current architecture assigns agents a compliance propensity and a social learning parameter, allowing simulation of norm diffusion and resistance under varying enforcement regimes.

Extending the SNMS model to include LSE would require three additions. First, a class of norms with the structural unenforceability property: norms for which the compliance cost parameter exceeds all agents' resource endowments. Second, a regulator agent with an observation-and-learning module that updates its enforcement targeting rule based on the correlation between prior enforcement actions and subsequent reductions in contestation activity. Third, a cross-domain contestation variable for each firm-agent, recording its activity across multiple regulatory domains and allowing the model to track whether enforcement in one domain suppresses activity in others.

With these additions, the SNMS could simulate the bug-to-feature transition under controlled parameter variation, including the enforcement differential, the political payoff phi, the initial regulatory complexity level, and the CLI score of the simulated institutional environment. The simulation would generate synthetic time series of contestation activity against which Predictions 2 and 3 of Section V could be calibrated. Prediction 3, on post-government-change persistence, is particularly amenable to simulation: the model can reset the regulator agent's beliefs at a specified time point and observe whether the equilibrium reconstitutes itself within the predicted 12-24 month window.

Epistemological Positioning

This paper is a contribution to a Lakatosian research program (Lakatos, 1978) whose hard core consists of three commitments: that legal norms are cultural replicators subject to evolutionary selection (Dawkins, 1976; Hull, 1988); that legal institutions are extended phenotypes of those replicators (Dawkins, 1982); and that evolutionary game theory provides the appropriate formalism for modeling institutional equilibria (Maynard Smith, 1982). The present paper operates in the protective belt: it introduces a specific mechanism, the lateral silencing exaptation, that is consistent with the hard core and extends its explanatory reach to a class of phenomena, persistently unenforceable norms, that prior versions of the program did not address.

The epistemological stance follows Popper (1959) on demarcation: the paper provides falsifiable predictions in Section V. It follows Campbell (1974) on evolutionary epistemology: the argument is treated as a hypothesis subject to selection by empirical evidence, not as a deductive derivation from first principles.

Limitations

Three limitations merit explicit acknowledgment. First, the Georgian case relies substantially on secondary documentation, including the 2006 Harvard Kennedy School case study and World Bank reports, rather than primary archival sources. The claim that Shevardnadze's inner circle deliberately identified the silencing property by 1995 is supported by these secondary sources but has not been verified against primary administrative records, which may not be accessible. The timing of intentionality should be treated as an estimate with significant uncertainty.

Second, the Argentine data on cross-domain suppression of contestation rests on publicly available parliamentary testimony records and the 2022 Congressional Research Service report. A proper test of Prediction 1 would require matched-pair comparison of firms in dual-control sectors versus comparable single-mode sectors, controlling for observable firm characteristics. That analysis is beyond the scope of this paper.

Third, the game-theoretic formalization in Section IV abstracts from several realistic complications: heterogeneous firms with different enforcement risk profiles; multi-round interactions where enforcement history affects beliefs; and the possibility of collective action by firms to overcome the silencing equilibrium. These extensions are not pursued here.

Conclusions

Structurally unenforceable norms do not survive because of legislative inertia or state incapacity, though both may be present. They survive, and specifically they survive across regime changes with heterogeneous policy preferences, because they develop a secondary functional property that creates a constituency for their maintenance. That property is the lateral suppression of political contestation by permanently non-compliant regulated agents.

The transition from unintended byproduct to deliberate governance instrument, from bug to feature, follows an evolutionary logic: once a regulatory actor observes the silencing effect, the payoff from maintaining the dysfunctional norm exceeds the payoff from correcting it. The equilibrium is stable, and it is stable precisely because the mechanism that would generate pressure for reform, contestation by affected agents, is the mechanism that the norm suppresses.

This is the deepest form of institutional lock-in: not the kind that requires constitutional barriers or judicial veto players, but the kind that disables the political actors who would otherwise demand reform. The lateral silencing exaptation is, in this sense, an institution that reproduces the conditions of its own persistence.

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AI Disclosure: AI tools were used as research support for locating sources, extracting information from academic literature, and reviewing drafts for consistency. All analysis, argumentation, and conclusions are the author's own. The concept of the Lateral Silencing Exaptation, the two-case typology, the game-theoretic formalization, and the connection to the EPT/EGT research program were conceived, structured, verified, and revised by the author. All substantive content is the author's original contribution.

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